



# Power Apps

Hands-on Project



# HR System

Employee Management System using  
SharePoint & Power Apps.



# SharePoint Capstone Project



## EMPLOYEE MANAGEMENT SYSTEM (CRUD APPLICATION) USING SHAREPOINT & POWER APPS



### Project Scenario

A medium-sized company currently manages employee records manually using spreadsheets, causing problems such as:

- ✗ Data redundancy
- ✗ Difficulty searching employee records
- ✗ Poor data organization
- ✗ Slow employee management process

The company wants a digital solution to:

- ✓ Store employee information centrally
- ✓ Add new employees
- ✓ Update employee information
- ✓ Delete resigned employees
- ✓ View employee details
- ✓ Search and filter employee records

You are hired as a **Power Platform Developer** to design and develop an **Employee Management System** using:

- **SharePoint Online** → Backend Database
- **Power Apps Canvas App** → User Interface



### Project Objectives

Students must develop an application that demonstrates:



#### SharePoint Integration

- ✓ Create SharePoint Lists
- ✓ Configure SharePoint columns
- ✓ Connect SharePoint to Power Apps



#### CRUD Operations

Students must implement:

- + **Create Employee**
- 👁️ **Read/View Employee**
- ✎️ **Update Employee**
- 🗑️ **Delete Employee**



#### User Interface Design

Students must design:

-  Home Screen
-  Employee List Screen
-  Add Employee Screen
-  Update Employee Screen
-  Employee Details Screen
-  Delete Confirmation Feature

### ✓ **Data Validation**

Students must implement:

-  Email validation
-  Phone number validation
-  Empty field prevention
-  Error messages

### **Module 1: SharePoint Backend Setup**

Students must create a **SharePoint List** named:

#### **EmployeeManagement**

#### **Required Columns**

|  <b>Column Name</b> |  <b>Data Type</b> |  <b>Description</b> |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
|  EmployeeID        | Single Line of Text                                                                                | Unique employee ID                                                                                   |
|  FullName         | Single Line of Text                                                                                | Employee full name                                                                                   |
|  Gender           | Choice                                                                                             | Male / Female                                                                                        |
|  Department       | Choice                                                                                             | HR, IT, Finance, Marketing                                                                           |
|  JobTitle         | Single Line of Text                                                                                | Employee role                                                                                        |
|  Email            | Single Line of Text                                                                                | Employee email                                                                                       |
|  PhoneNumber      | Text                                                                                               | Contact number                                                                                       |
|  HireDate         | Date                                                                                               | Employment start date                                                                                |
|  Salary           | Currency                                                                                           | Monthly salary                                                                                       |
|  Status           | Choice                                                                                             | Active / Inactive                                                                                    |
|  Address          | Multiple Lines                                                                                     | Employee address                                                                                     |
|  ProfilePhoto     | Image (Optional)                                                                                   | Employee image                                                                                       |

### **Deliverable**

-  Screenshot of SharePoint List Structure

## **Module 2: Home Screen**

Students must design a professional **Home Page Dashboard**.

### **Required Components**

#### **Navigation Menu**

Buttons for:

-  View Employees
-  Add Employee
-  Update Employee
-  Delete Employee
-  Exit / Logout

#### **Dashboard Cards**

Display:

-  Total Employees
-  Active Employees
-  Inactive Employees

#### **Footer**

 Student Name

 Project Title

#### **Deliverable**

 Screenshot of Home Screen

## **Module 3: Employee List (Read Operation)**

Students must create an **Employee List Page**.

### **Required Features**

#### **Gallery Control**

Display:

-  Employee ID
-  Employee Name
-  Department
-  Job Title
-  Status

#### **Search Functionality**

Search using:

-  Employee Name
-  Employee ID
-  Department

#### **Filter Functionality**

Filter by:

-  Department
-  Status

#### **Sorting**

- ↑ Ascending
  - ↓ Descending
  - 🎯 **Action Buttons**
  - 👁️ View Details
  - ✏️ Edit Employee
  - 🗑️ Delete Employee
  - 🏠 Back to Home
  - 🎯 **Deliverable**
  - 📷 Employee List Page Screenshot
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## + **Module 4: Add Employee (Create Operation)**

Students must create an **Add Employee Screen**.

### **Required Fields**

- ID Employee ID
- 👤 Full Name
- ♂️ Gender
- 🏢 Department
- 💼 Job Title
- ✉️ Email
- 📞 Phone Number
- 📅 Hire Date
- 💰 Salary
- 📄 Status
- 🏠 Address

### ○ **Required Buttons**

- 💾 Save
- ↺️ Reset
- ✖️ Cancel

### ✓ **Validation Rules**

- 🚫 Required fields cannot be empty
- ✉️ Email Validation
- 📞 Phone number validation
- ⚠️ Show error messages for invalid input

### 🎯 **Deliverable**

- 📷 Add Employee Form Screenshot
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## 👤 **Module 5: Employee Details Screen**

Students must create a detailed employee information page.

### **Display Information**

-  Employee Photo
-  Full Name
-  Employee ID
-  Department
-  Job Title
-  Salary
-  Contact Information
-  Status
-  Address

### Required Buttons

-  Edit
-  Delete
-  Back

### Deliverable

-  Employee Details Screen Screenshot

## Module 6: Update Employee (Update Operation)

Students must create an **Edit Employee Screen**.

### Requirements

-  Auto-fill selected employee data
-  Allow editing existing information
-  Save updated changes

### Success Message

-  Employee Updated Successfully

### Deliverable

-  Update Employee Page Screenshot

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## Module 7: Delete Employee (Delete Operation)

Students must implement employee deletion.

### Requirements

-  Show confirmation before deletion

Example Message:

 **Are you sure you want to delete this employee?**

### Success Message

-  Employee Deleted Successfully

### Deliverable

-  Delete Functionality Screenshot

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## Module 8: Navigation

Students must implement navigation between screens.

### Navigation Requirements

-  Home → Employee List
-  Home → Add Employee
-  Employee List → Details
-  Details → Edit
-  Edit → Save & Return
-  **Deliverable**
-  Navigation Demonstration Screenshot

## **Module 9: App Design Requirements**

Students must ensure:

### **Professional Design**

-  Consistent colors
-  Proper alignment
-  Appropriate icons
-  Responsive design

### **User Experience**

-  Easy navigation
-  Clear buttons
-  User-friendly forms

### **Error Handling**

Meaningful validation messages

### **Deliverable**

-  Final App Design Screenshot

## **Step-by-Step Implementation Guide**

### **Phase 1: Requirement Analysis**

-  Understand requirements
  -  Identify system users
  -  Design application screens
-  **Deliverable:** Requirement Document

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### **Phase 2: SharePoint Setup**

-  Create SharePoint Site
  -  Create **EmployeeManagement** list
  -  Configure columns
  -  Add sample employee data
-  **Deliverable:** SharePoint Screenshot

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### **Phase 3: Power Apps Setup**

- 1 Create Canvas App
  - 2 Connect SharePoint Data Source
  - 3 Apply theme & layout
-  **Deliverable:** Data Connection Screenshot
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#### **Phase 4: UI Design**

- 1 Design Home Screen 
- 2 Design CRUD Pages 
- 3 Add Navigation 

 **Deliverable:** UI Screenshots

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#### **Phase 5: CRUD Implementation**

-  Create Employee
-  Read Employee
-  Update Employee
-  Delete Employee

 **Deliverable:** Functional CRUD App

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#### **Phase 6: Testing**

Students must test:

-  Add Employee
-  Search Employee
-  Update Employee
-  Delete Employee
-  Invalid Email Validation

 **Deliverable:** Testing Report

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#### **Final Submission Requirements**

Students must submit:

-  **Power Apps Export File (.msapp)**
-  **SharePoint List Screenshot**
-  **Project Report (10–15 Pages)**
-  **Presentation Slides (10–15 Slides)**
-  **Demo Video (5–10 Minutes)**



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**First solve the  
problem. Then,  
write the code.**

**- John Johnson**

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